VIJAY P

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PROFILE

Full-stack and AI/ML-focused Computer Science undergraduate building secure, data-driven systems end to end, from federated edge intelligence and explainable ML to blockchain identity and IoT-attested secure messaging.

EDUCATION

B.Tech in Computer Science and Engineering, Vellore Institute of Technology, Chennai 2023 – 2027
CGPA: **8.57/10** Class XII: **95.2%** Class X: **96.4%**

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, C++, C, Java, SQL, Solidity
Frontend:	React.js, React Router, HTML5, CSS3, Tailwind CSS, Vite
Backend:	FastAPI, Node.js, Express.js, RESTful APIs, WebSockets, Pydantic, Mongoose
Machine Learning & AI:	scikit-learn, LightGBM, Pandas, NumPy, SHAP, federated learning, LLMs, RAG
Security & Blockchain:	RSA, AES-256, SHA-256, bcrypt, TOTP MFA, IEEE 1609.2 PKI, Hardhat, OpenZeppelin
Databases & DevOps:	MongoDB, SQLite, PostgreSQL, Docker, Docker Compose, Git/GitHub, pytest, Vitest, Linux

PROJECTS

Decentralized V2X Ecosystem | TypeScript, React, FastAPI, scikit-learn, WebSockets GitHub

- Built a decentralized vehicle-to-everything system with peer-to-peer hazard sharing over **6 ETSI C-ITS message types**, roadside-unit edge inference, a city digital twin, and IEEE 1609.2 certificates with pseudonym rotation.
- Against a centralized baseline over **5 seeds** with 95% CIs: cut cloud uplink traffic by **88.7%**, kept **100% availability** through a cloud outage (vs. 0%), and raised hazard-detection precision from **0.861 to 0.994**.
- Trained a regional congestion model with **federated averaging** (**24 rounds**, **275.6 KB** of raw telemetry kept local) and shipped a **7-scenario** guided demo backed by **360+ automated tests** (pytest + Vitest).

Diabetes 30-Day Readmission Prediction | Python, LightGBM, SHAP, FastAPI, Streamlit GitHub

- Built an end-to-end ML pipeline on the UCI Diabetes 130-US Hospitals dataset (**101,766 encounters**, **50+ features**) with feature engineering, stratified train/validation/test splits, and class-imbalance handling.
- Benchmarked LightGBM against logistic regression, reaching **0.667 ROC-AUC** and **0.228 PR-AUC** on the held-out test set, a **4.1%** and **9.6%** relative lift over the baseline (0.641 / 0.208).
- Added global and patient-level **SHAP** explanations, threshold tuning, and fairness analysis; served single and batch inference through **FastAPI** and a **Streamlit** dashboard with risk buckets and recommendations.

TourSecure – Intelligent Tourist Safety Platform | React, Node.js, FastAPI, MongoDB, Solidity, Docker GitHub

- Built a multi-service safety platform for **3 user roles** (tourists, authorities, responders) with a **7-tab** React/TypeScript UI covering SOS alerts, safety heatmaps, itinerary monitoring, e-FIR filing, and digital ID.
- Implemented polygon and distance-based geofencing plus route-deviation and inactivity checks as **2 independent FastAPI microservices**, with extension points for learned anomaly-detection models.
- Deployed an **ERC-721** TouristID smart contract (Hardhat, OpenZeppelin access control) and shipped the full stack as a reproducible **6-service Docker Compose** deployment.

PhysiLock – IoT-Attested Secure Messaging Platform | Python, FastAPI, cryptography, SQLite, TOTP GitHub

- Developed a secure messaging and file-transfer platform with **20 REST endpoints**, using **RSA-2048** (PSS signatures, OAEP key wrapping) and **AES-256** hybrid encryption with SHA-256 integrity checks.
- Engineered a context-aware risk engine scoring **4 signals** (device, IP, login time, IoT attestation) into a **3-tier decision**: allow, require TOTP, or block.
- Built an IoT device simulator for hardware attestation, with bcrypt password hashing and audit logging.

ACHIEVEMENTS & LEADERSHIP

- **Institute Finalist, Smart India Hackathon 2025:** Built TourSecure for tourist risk assessment and incident response.
- **First Place, Inter-College Web Development Competition 2024:** Built a full-stack MERN e-commerce platform.
- **Event Coordinator, College Club (2024 – Present):** Plan and run technical and cultural events with cross-functional teams.

CERTIFICATIONS & LANGUAGES

- **IBM Generative AI Advanced Certification** – IBM
- **Languages:** English, Tamil, Hindi